

## ASHEBORO, NC, USA

WMO: 746935

Lat: 35.654N

Lon: 79.895W

Elev: 205

StdP: 98.89

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 03709

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.2       | -4.7 | -18.0                         | 0.8 | -1.6 | -15.6 | 1.0 | -0.7 | 8.3                   | 13.6 | 7.5 | 12.1 | 1.2                   | 350  | 0.363 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 10.0                      | 33.9            | 23.1 | 32.7 | 23.3 | 32.0 | 23.2 | 25.7                | 29.8 | 24.9 | 29.7 | 24.3 | 29.0 | 2.2                     | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.2                            | 19.6 | 27.4 | 23.6 | 18.9 | 26.9 | 22.9 | 18.1 | 26.2 | 80.2          | 30.2 | 76.6 | 29.9 | 74.5 | 29.0 | 31.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 7.2               | 5.8  | 5.1 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -10.1                      | 35.8 | 2.8                | 1.8 | -12.1                                              | 37.1 | -13.7      | 38.1 | -15.2      | 39.1 | -17.2      | 40.4 |  |  |
|                   |      |     | -11.3                      | 26.9 | 2.6                | 1.5 | -13.2                                              | 28.0 | -14.8      | 28.9 | -16.3      | 29.7 | -18.2      | 30.8 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.9   | 5.1   | 6.8   | 10.7  | 16.0  | 20.1  | 24.2  | 26.0  | 25.3  | 22.1  | 16.3  | 10.7  | 7.3  |
|                                                                                  | DBStd       | 8.51   | 5.86  | 5.20  | 5.40  | 4.59  | 3.80  | 2.85  | 2.19  | 2.50  | 3.12  | 4.51  | 4.87  | 5.10 |
|                                                                                  | HDD10.0     | 526    | 173   | 117   | 59    | 7     | 0     | 0     | 0     | 0     | 0     | 4     | 49    | 116  |
|                                                                                  | HDD18.3     | 1781   | 411   | 325   | 241   | 98    | 28    | 1     | 0     | 0     | 7     | 94    | 234   | 342  |
|                                                                                  | CDD10.0     | 2692   | 22    | 27    | 81    | 186   | 314   | 427   | 496   | 475   | 363   | 200   | 69    | 33   |
|                                                                                  | CDD18.3     | 905    | 1     | 1     | 5     | 27    | 84    | 178   | 237   | 217   | 120   | 31    | 4     | 1    |
|                                                                                  | CDH23.3     | 7677   | 2     | 4     | 41    | 248   | 654   | 1575  | 2174  | 1869  | 872   | 209   | 26    | 3    |
|                                                                                  | CDH26.7     | 2806   | 0     | 0     | 3     | 49    | 171   | 613   | 906   | 747   | 273   | 43    | 2     | 0    |
| Wind                                                                             | WSAvg       | 1.7    | 2.0   | 2.1   | 2.2   | 2.3   | 1.8   | 1.4   | 1.3   | 1.1   | 1.3   | 1.4   | 1.6   | 1.7  |
| Precipitation                                                                    | PrecAvg     | 1150   | 88    | 79    | 107   | 92    | 87    | 103   | 120   | 109   | 111   | 93    | 86    | 81   |
|                                                                                  | PrecMax     | 1565   | 144   | 150   | 210   | 190   | 216   | 233   | 242   | 202   | 266   | 338   | 209   | 147  |
|                                                                                  | PrecMin     | 829    | 29    | 32    | 41    | 19    | 26    | 20    | 53    | 42    | 15    | 0     | 15    | 39   |
|                                                                                  | PrecStd     | 170    | 37    | 31    | 44    | 45    | 46    | 56    | 50    | 37    | 72    | 79    | 50    | 32   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 22.0  | 23.8  | 27.2  | 30.9  | 32.2  | 35.0  | 36.0  | 36.2  | 32.9  | 30.2  | 26.3  | 22.9 |
|                                                                                  |             | MCWB   | 16.9  | 16.6  | 16.3  | 17.7  | 21.6  | 22.6  | 23.2  | 23.2  | 21.5  | 22.2  | 18.2  | 17.4 |
|                                                                                  | 2%          | DB     | 18.9  | 21.0  | 24.1  | 27.9  | 30.2  | 32.9  | 33.8  | 33.8  | 31.3  | 27.6  | 23.0  | 20.4 |
|                                                                                  |             | MCWB   | 14.8  | 14.5  | 15.8  | 18.1  | 20.6  | 23.2  | 23.6  | 23.2  | 21.4  | 20.0  | 16.8  | 17.1 |
|                                                                                  | 5%          | DB     | 17.2  | 18.0  | 22.4  | 26.4  | 28.6  | 32.1  | 32.7  | 32.4  | 29.8  | 25.9  | 21.4  | 18.0 |
|                                                                                  |             | MCWB   | 13.3  | 13.0  | 14.5  | 17.1  | 20.3  | 22.9  | 23.9  | 23.6  | 21.5  | 19.2  | 16.0  | 15.0 |
|                                                                                  | 10%         | DB     | 14.5  | 15.9  | 19.9  | 24.1  | 27.3  | 30.3  | 31.5  | 30.9  | 27.9  | 23.6  | 19.0  | 16.2 |
|                                                                                  |             | MCWB   | 10.4  | 10.9  | 13.8  | 16.6  | 19.5  | 22.4  | 23.6  | 23.2  | 21.1  | 19.0  | 14.8  | 12.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 18.0  | 18.2  | 19.3  | 22.4  | 24.0  | 27.9  | 28.4  | 27.8  | 25.2  | 23.6  | 21.4  | 19.6 |
|                                                                                  |             | MCDB   | 20.4  | 21.5  | 21.9  | 25.9  | 28.1  | 29.5  | 31.3  | 30.0  | 28.7  | 26.2  | 24.5  | 20.9 |
|                                                                                  | 2%          | WB     | 16.1  | 16.6  | 17.4  | 20.3  | 22.3  | 25.1  | 25.6  | 25.3  | 24.1  | 22.6  | 19.0  | 17.8 |
|                                                                                  |             | MCDB   | 18.3  | 19.2  | 21.5  | 24.7  | 26.6  | 29.2  | 30.5  | 30.2  | 27.7  | 25.3  | 21.3  | 19.6 |
|                                                                                  | 5%          | WB     | 14.1  | 14.4  | 16.3  | 19.2  | 21.6  | 24.1  | 24.9  | 24.8  | 23.3  | 21.3  | 17.4  | 15.9 |
|                                                                                  |             | MCDB   | 16.6  | 17.6  | 20.2  | 23.8  | 25.9  | 29.1  | 30.1  | 29.5  | 26.8  | 24.3  | 20.0  | 17.8 |
|                                                                                  | 10%         | WB     | 11.4  | 11.8  | 15.1  | 17.9  | 20.8  | 23.4  | 24.3  | 24.2  | 22.4  | 19.7  | 15.2  | 13.4 |
|                                                                                  |             | MCDB   | 13.8  | 14.5  | 19.0  | 22.1  | 25.2  | 28.2  | 29.2  | 28.6  | 25.8  | 22.9  | 18.6  | 15.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.2  | 10.7  | 11.6  | 12.0  | 11.0  | 10.4  | 10.0  | 9.8   | 10.1  | 11.1  | 12.0  | 10.3 |
|                                                                                  |             | MCDBR  | 12.2  | 13.3  | 13.2  | 14.0  | 12.5  | 11.8  | 11.4  | 11.6  | 12.6  | 12.4  | 13.6  | 11.6 |
|                                                                                  |             | MCWBR  | 9.0   | 8.5   | 7.0   | 6.7   | 5.3   | 4.5   | 4.4   | 4.1   | 4.4   | 6.2   | 8.5   | 8.2  |
|                                                                                  | 5% WB       | MCDBR  | 10.7  | 11.7  | 11.8  | 12.2  | 10.4  | 10.5  | 10.3  | 9.7   | 9.3   | 10.0  | 10.9  | 9.7  |
|                                                                                  | MCWBR       | 9.2    | 9.3   | 8.3   | 7.5   | 5.2   | 5.3   | 4.7   | 4.5   | 5.1   | 6.0   | 8.7   | 8.0   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.311  | 0.324 | 0.359 | 0.398 | 0.449 | 0.484 | 0.533 | 0.513 | 0.422 | 0.383 | 0.328 | 0.310 |      |
|                                                                                  | taud        | 2.504  | 2.460 | 2.397 | 2.288 | 2.182 | 2.120 | 2.006 | 2.047 | 2.296 | 2.376 | 2.495 | 2.536 |      |
|                                                                                  | Ebn at Noon | 893    | 919   | 910   | 886   | 842   | 809   | 766   | 773   | 834   | 837   | 863   | 871   |      |
|                                                                                  | Edh at Noon | 82     | 96    | 111   | 131   | 148   | 157   | 175   | 165   | 122   | 103   | 82    | 75    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.59   | 3.25  | 4.31  | 5.39  | 5.98  | 6.39  | 5.98  | 5.45  | 4.58  | 3.74  | 2.85  | 2.22  |      |
|                                                                                  | RadStd      | 0.17   | 0.40  | 0.30  | 0.41  | 0.49  | 0.47  | 0.39  | 0.26  | 0.41  | 0.49  | 0.23  | 0.19  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (66 neighbors) | +0.42 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +147    | +104    |

Nomenclature: See separate page